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# CATANBENCH: A REAL-TIME BENCHMARK FOR STRATEGIC REASONING IN SETTLERS OF CATAN

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## ABSTRACT

We present CatanBench, a benchmark for evaluating strategic decision making in the board game Settlers of Catan. Unlike benchmarks that primarily score agents by final game outcome, CatanBench evaluates individual decisions at the moment they are made. The benchmark covers short-term tactical moves, medium-term resource and build prioritization, and long-term strategic commitments such as pursuing Longest Road or Largest Army. Each decision is compared against the best legal alternative available in the current game state using option-ratio and component-weighted rubrics. CatanBench is implemented inside a complete Node.js/React Catan environment with a Socket.IO multiplayer backend and a Python tool API for agents. During live gameplay, agent actions automatically trigger task evaluators, persist task scores, and support both real-time telemetry and offline aggregate analysis.

## 1 INTRODUCTION

Many game benchmarks evaluate whether an agent wins, but this is often too sparse to diagnose strategic reasoning. In Settlers of Catan, a final win or loss is shaped by hidden information, stochastic dice rolls, turn order, opponent negotiation, and long delayed consequences. An agent can make a strong settlement placement and still lose to unlucky dice; it can also win after repeatedly making fragile trades because opponents failed to punish them. This makes outcome-only evaluation a poor fit for studying whether an agent understands the quality of the choices available at each decision point.

CatanBench addresses this problem by evaluating decisions during live gameplay. The benchmark covers reasoning across short-, medium-, and long-term strategic moves. Short-term tasks include tactical decisions such as where to place a road, which opponent to rob, and whether to accept a trade. Medium-term tasks evaluate turn-level prioritization, including whether to build now or save, whether to buy a development card, and how to discard after a seven. Long-term tasks evaluate strategic commitments such as opening settlement placement, pursuing Longest Road, and pursuing Largest Army. Each task scores the chosen action against the best legal alternative currently available, rather than against a fixed answer that ignores the board state.

Existing game and agent benchmarks are insufficient for this setting in three ways. First, many board-game environments emphasize outcome, search strength, or generic reinforcement-learning performance, while CatanBench targets the intermediate decisions that make a strategy coherent over time. Second, Catan requires simultaneous management of public board geometry, private hands, stochastic dice, and open-ended negotiation; benchmarks that isolate perfect-information search or single-step planning miss this mixed-information structure. Third, language-agent benchmarks often evaluate final textual answers, while CatanBench evaluates tool-mediated actions embedded in a changing multiplayer environment.

Our contribution is a 20-task benchmark for Catan decision making that runs during real games. The benchmark evaluates each decision against the legal alternatives available at that exact moment of gameplay using deterministic heuristic rubrics. It records both task-level quality and game-level performance, enabling analysis of whether agents win, how they win, and which decision families cause failure. CatanBench runs on a complete Node.js/React implementation of Settlers of Catan with a Socket.IO multiplayer backend. Agents interact through a Python tool API. Each agent action

054 automatically triggers the relevant evaluator, stores the score, and contributes to real-time telemetry  
055 and offline aggregate reports.

## 057 2 RELATED WORK

059 CatanBench is related to game-playing benchmarks, multi-agent environments, and benchmarks  
060 for negotiation or hidden-information reasoning. The Arcade Learning Environment evaluates  
061 reinforcement-learning agents across Atari games and established the value of standardized game  
062 interfaces and comparable scores (Bellemare et al., 2013). OpenSpiel provides a broad collection of  
063 environments and algorithms for reinforcement learning and planning in games, including perfect-  
064 and imperfect-information settings (Lancot et al., 2019). The Hanabi Challenge focuses on coop-  
065 erative imperfect-information play and theory-of-mind reasoning under hidden hands (Bard et al.,  
066 2020). Diplomacy research environments study negotiation, long-horizon planning, and multi-agent  
067 commitments in a strategic board game with natural-language or structured communication (Paque-  
068 tte et al., 2019).

069 CatanBench differs from these benchmarks by scoring decision quality during a full game instead  
070 of relying primarily on final outcome. The benchmark is not only a Catan environment; it is an  
071 instrumentation layer over gameplay that records whether an agent’s selected legal action is close  
072 to the best legal alternative under a task-specific rubric. This fills a gap between outcome-based  
073 game benchmarks and static reasoning benchmarks: it evaluates situated strategic judgment under  
074 changing board state, stochastic dice outcomes, hidden opponent resources, and negotiation.

## 076 3 BENCHMARK DESIGN

### 078 3.1 TASK DOMAIN AND SCOPE

080 CatanBench covers four categories of Catan decisions. Spatial placement tasks evaluate settlement  
081 and road locations. Resource management tasks evaluate trading, discarding, bank trades, and build-  
082 versus-save decisions. Opponent interaction tasks evaluate robber placement, victim selection, trade  
083 targeting, counteroffers, and leader blocking. Strategic commitment tasks evaluate whether an agent  
084 should pursue Longest Road or Largest Army over a delayed horizon.

085 The benchmark is designed to test decision quality rather than only game outcome. In scope are  
086 individual legal decisions made during real gameplay, their task-specific scores, pass/fail thresholds,  
087 and aggregate performance over games. Out of scope are natural-language explanation grading,  
088 subjective human preference scoring, and exhaustive game-theoretic solution of Catan. We do not  
089 claim that the heuristic best action is a solved optimal policy. Instead, CatanBench follows an option-  
090 ratio benchmark framework: enumerate or construct legal alternatives available at the decision point,  
091 score each alternative with a deterministic rubric, and score the selected action as a ratio against the  
092 best legal alternative.

### 093 3.2 TASK SUITE

095 Table 1 summarizes the 20 implemented tasks. The suite contains 12 easy tasks, 5 medium tasks,  
096 and 3 hard tasks.

### 098 3.3 REPRESENTATIVE TASKS

099 **Initial settlement location selection.** The input is a setup-phase game-state snapshot containing  
100 the board layout, legal settlement vertices, adjacent hex resources and dice numbers, port locations,  
101 occupied vertices, and legal outgoing road options. The output is a legal settlement vertex. Each  
102 candidate is scored as:

$$104 \text{score} = 0.40 \cdot \text{production} + 0.25 \cdot \text{diversity} + 0.20 \cdot \text{expansionAccess} + 0.15 \cdot \text{portSynergy}. \quad (1)$$

106 Production uses Catan dice probability weights; diversity measures distinct adjacent resource types;  
107 expansion access counts future legal outgoing road choices; and port synergy rewards 2:1 or 3:1 port  
access. The selected vertex is compared against the best legal vertex. The current pass threshold is

| Task ID                                   | Category       | Difficulty | Status      |
|-------------------------------------------|----------------|------------|-------------|
| initial-settlement-location-selection     | Spatial        | Hard       | Implemented |
| settlement-location-selection             | Spatial        | Easy       | Implemented |
| road-placement-direction                  | Spatial        | Easy       | Implemented |
| knight-card-playing-decision              | Tactical       | Easy       | Implemented |
| year-of-plenty-card-playing-decision      | Tactical       | Easy       | Implemented |
| monopoly-card-playing-decision            | Tactical       | Easy       | Implemented |
| road-building-card-playing-decision       | Tactical       | Easy       | Implemented |
| robber-placement                          | Tactical       | Easy       | Implemented |
| robber-victim-selection                   | Tactical       | Easy       | Implemented |
| accept-or-reject-trade-offers             | Negotiation    | Easy       | Implemented |
| generate-trade-offers                     | Negotiation    | Easy       | Implemented |
| select-targeted-trade-partner             | Negotiation    | Easy       | Implemented |
| generate-counter-trade-offer              | Negotiation    | Easy       | Implemented |
| build-vs-save-decision                    | Prioritization | Medium     | Implemented |
| city-vs-settlement-vs-road-prioritization | Prioritization | Medium     | Implemented |
| development-card-purchase-decision        | Prioritization | Medium     | Implemented |
| discard-strategy-after-seven              | Prioritization | Medium     | Implemented |
| bank-trade-decision                       | Prioritization | Medium     | Implemented |
| decide-pursue-longest-road                | Strategic      | Hard       | Implemented |
| decide-pursue-largest-army                | Strategic      | Hard       | Implemented |

Table 1: CatanBench task suite. Difficulty reflects the horizon and planning complexity required by the decision, not only implementation complexity.

0.75 because opening placements are highly consequential but multiple strong starts can be defensible. This task is challenging because the first settlement shapes resource income, trade dependence, expansion paths, and the feasibility of future ports before the agent has observed any dice outcomes.

**Discard strategy after seven.** The input is a player hand after a seven is rolled, the required discard count, the current board state, legal future build plans, and resource need/surplus estimates. The output is a legal discard bundle. The evaluator enumerates legal discard bundles and scores the retained hand:

$$\text{retainedHandValue} = 0.55 \cdot \text{bestAffordable} + 0.30 \cdot \text{bestFuture} + 0.15 \cdot \text{diversity}. \quad (2)$$

The final score is  $\text{RetainedHandValue} / \text{BestRetainedValue}$ , and the pass threshold is 0.95. This task is challenging because the agent must reason about opportunity cost: the discarded cards matter less than whether the remaining hand preserves an immediate build, a near-term plan, and enough diversity to avoid bottlenecks.

**Pursue Longest Road.** The input is a mid- or late-game snapshot containing road lengths, visible scores, current Longest Road holder, opponent road rival, resource commitments, and later outcome telemetry. The output is a binary strategic commitment: pursue or defer. Unlike immediate tasks, this task creates a pending episode and finalizes after a four-turn horizon, decisive resolution, or game end. The delayed score combines blocking effectiveness, opponent road threat, self road synergy, and resource cost. The pass threshold is 0.95. This task is challenging because the correct decision is not a single road placement; it is a multi-turn commitment whose value depends on opponent progress, available resources, and whether the investment produces victory-point leverage.

### 3.4 TASK DESIGN PRINCIPLES

The central design principle is comparison against the best legal alternative. This avoids treating a reasonable but second-best move as fully wrong and makes scores robust to the current board state. We use component-weighted rubrics when the decision has interpretable factors. For example, settlement placement assigns 0.40 weight to production because early income determines the opening trajectory, and 0.20 to expansion access because path-dependent road investments are difficult to recover from if the settlement is geometrically constrained.

Coverage was achieved by mapping the benchmark to the major action families a Catan player can take: spatial placement, card play, negotiation and trading, build prioritization, discarding, and

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strategic commitment. Difficulty was calibrated by horizon. Easy tasks are immediate tactical decisions with direct legal alternatives. Medium tasks require turn-level prioritization across resources and possible plans. Hard tasks require opening or multi-turn strategic reasoning, where immediate scoring can be misleading and delayed evaluation is needed.

## 4 EVALUATION PROTOCOL

Agents are evaluated in benchmark-enabled Catan games. Each action is recorded with metadata including benchmark ID, agent ID, agent version, map layout, opponent policy set, scenario tags, selected option, task score, task threshold, latency, and legality. Task scores are computed as either an option ratio against the best legal alternative, a component-weighted score normalized to  $[0, 1]$ , or a retained-hand ratio for discarding. A task succeeds when its score exceeds the task-specific threshold.

The benchmark reports nine weighted metrics: win rate, average final victory points, average rounds to win, task success rate, average latency per turn, illegal move rate, robustness, consistency, and generalization. The default metric weights are 0.20 for win rate, 0.10 for average final victory points, 0.10 for average rounds to win, 0.20 for task success rate, 0.075 for average latency per turn, 0.075 for illegal move rate, 0.075 for robustness, 0.075 for consistency, and 0.10 for generalization. Primary metrics combine gameplay, reasoning, and operational performance. Secondary metrics aggregate across scenario slices such as resource scarcity, alternate maps, seat variation, and opponent-policy variation.

Reports should include overall score, task success rate, average task score, per-task breakdowns, win rate, average final victory points, average rounds to win, illegal move rate, latency per turn, and slice-level robustness/generalization. We also recommend reporting confidence intervals over multiple games and separate breakdowns by task difficulty.

## 5 BENCHMARK VALIDATION

We conducted pilot testing by having humans play complete games while CatanBench logged tasks in real time. These games were used to sanity-check whether task triggers occurred at the intended decision points and whether the heuristic rankings aligned with intuitive Catan play. For example, the settlement rubric ranked high-production, diverse, expandable locations highly, and the robber placement heuristic generally matched common denial strategies that target the leader while avoiding self-harm.

Task quality is supported by deterministic task specifications. Each of the 20 tasks has an unambiguous input snapshot, a discrete legal action space, and a defined scoring function. Shared heuristic building blocks such as production probability weights, resource diversity, expansion access, port synergy, build-plan values, and resource need/surplus are reused consistently across tasks. For long-horizon tasks, we moved from immediate scoring to delayed four-to-six-turn episodes so that strategic commitments are not penalized before their consequences can materialize.

We have not yet completed a formal inter-rater reliability study because CatanBench does not depend on human labels at evaluation time. Future validation should ask experienced players to rank sampled legal alternatives and measure agreement with the benchmark’s deterministic ordering. Preliminary human pilot results suggest that reasonable human play clears the pass threshold across task families, but a larger baseline-agent and human study is needed before making stronger claims.

## 6 CONCLUSION

CatanBench is a real-time benchmark for strategic reasoning in Settlers of Catan. It evaluates 20 live gameplay tasks across tactical decisions, resource prioritization, negotiation, and long-horizon commitments. By scoring decisions against the best legal alternative at the moment of play, CatanBench provides denser and more diagnostic feedback than final win rate alone. We expect the benchmark to help researchers study agents that must reason under hidden information, stochastic outcomes, negotiation, and delayed strategic consequences. Future work includes larger human and agent baselines, stronger negotiation modeling, additional scenario suites, and calibration against expert Catan play.

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## A BENCHMARK TASK EXPLANATIONS

### A.1 DESIGN PRINCIPLES

The benchmark uses three families of tasks:

- **Short-term tasks:** immediate tactical choices with direct board consequences.
- **Medium-term tasks:** action prioritization and resource conversion decisions.
- **Long-term tasks:** strategic race commitments whose value unfolds over several turns.

The main design goal is to compare decisions with the best legal alternative available at that moment (based on a heuristic approach), while keeping a threshold value to determine success or failure on the individual task. This system rewards near-optimal play when several choices are reasonable, and avoids placing fail labels on decisions that were reasonable but not perfectly aligned with the benchmark heuristic.

### A.2 SHARED HEURISTIC BUILDING BLOCKS

#### A.2.1 PRODUCTION EXPECTANCY

Some dice numbers have a higher probability of being rolled than others. Specifically, the probability weights are:

$$2, 12 \mapsto 1, \quad 3, 11 \mapsto 2, \quad 4, 10 \mapsto 3, \quad 5, 9 \mapsto 4, \quad 6, 8 \mapsto 5.$$

7 is a number used for the robber and does not produce resources. To compute the expected value of the resources gained on the next turn from a current settlement vertex,

$$\begin{aligned} \text{productionWeight} &= \sum \text{adjacent\_hex\_probability\_weights}, \\ \text{resourceProductionExpectancy} &= \min \left( 1, \frac{\text{productionWeight}}{15} \right). \end{aligned}$$

The denominator 15 is the theoretical maximum because a vertex touches at most three hexes and the maximum per-hex weight is 5.

#### A.2.2 RESOURCE DIVERSITY

Each settlement vertex is adjacent to up to three hexes, which can produce up to three different resource types (`distinct_adjacent_resource_types`).

Resource diversity is defined as

$$\text{resourceDiversity} = \min \left( 1, \frac{\text{distinct\_adjacent\_resource\_types}}{3} \right).$$

It matters because diverse starts reduce dependency on trading, improve build robustness, and lower the risk of bottlenecking on a single missing resource.

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### A.2.3 EXPANSION ACCESS

In crowded boards, some settlement locations have more future legal road placements than others, which translates into better expansion potential and better survivability in case of unfavorable dice rolls or robber placements. For a settlement candidate, the evaluator computes incident edges, keeps only edges where a road could legally be placed, and defines

$$\text{expansionAccess} = \min\left(1, \frac{\text{legal\_outgoing\_edges}}{3}\right).$$

### A.2.4 PORT SYNERGY

We use a discrete port heuristic:

- specific 2:1 port  $\rightarrow 1.0$
- generic 3:1 port  $\rightarrow 0.7$
- no port  $\rightarrow 0.0$

Ports matter because they convert surplus resources into missing ones; specific ports are usually stronger due to higher exchange efficiency.

### A.2.5 BUILD PLAN BASE SCORES

Several tasks construct plan candidates for road, settlement, city, and devCard. Each plan has a `baseScore`: the current heuristic value of that build family before applying affordability or distance-to-completion discounts. The shared definitions are:

- `road`: the maximum score among currently legal road placements under the road-placement rubric from Section A.3.3.
- `settlement`: the maximum score among currently legal settlement placements under the settlement rubric from Section A.3.2.
- `city`: the maximum score among legal city upgrades using  $0.65 \cdot \text{production} + 0.10 \cdot \text{diversity} + 0.05 \cdot \text{portSynergy} + 0.20$ .
- `devCard`: 0 if the development-card deck is unavailable; otherwise  $\max(0.2, 0.55 \cdot (0.5 + 0.5 \cdot \text{cityPlanScore}))$ , where `cityPlanScore` is the current best city-upgrade score computed with the city formula in this list.

### A.2.6 RESOURCE NEED AND SURPLUS

Resource need and surplus are used in trade and bank-trade scoring.

For each candidate plan and resource  $r$ , `cost[r]` is the number of cards of resource  $r$  required by the plan, `hand[r]` is the player's current count of resource  $r$ , and

$$\text{missing}[r] = \max(0, \text{cost}[r] - \text{hand}[r]).$$

`totalMissing` is the total number of resource cards still needed for the plan:

$$\text{totalMissing} = \sum_r \text{missing}[r].$$

Each plan is weighted according to how many resources are needed to achieve the plan:

$$\text{planWeight} = \frac{\text{baseScore}}{1 + 0.75 \cdot \text{totalMissing}},$$

where `baseScore` uses the shared build-plan definition from Section A.2.5. For each resource  $r$ ,

$$\text{need}[r] = \text{planWeight} \cdot \frac{\text{missing}[r]}{\max(1, \text{totalMissing})}.$$

For surplus, the evaluator reserves part of the current hand for weighted plans:

$$\text{reserved}[r] = \min(\text{hand}[r], \sum \text{planWeight} \cdot \text{cost}[r]),$$

then treats any unreserved cards of resource  $r$  as surplus. The surplus map is normalized by the largest unreserved resource count in the hand:

$$\text{surplus}[r] = \frac{\max(0, \text{hand}[r] - \text{reserved}[r])}{\max(1, \max_s \max(0, \text{hand}[s] - \text{reserved}[s]))},$$

where  $s$  ranges over all resource types. This avoids earlier failure modes where a resource such as ore looked important merely because it was scarce in hand, even when the board strongly favored early road and settlement expansion.

### A.3 TASK-BY-TASK EXPLANATIONS

#### A.3.1 INITIAL-SETTLEMENT-LOCATION-SELECTION

$\text{Score} = 0.40 \cdot \text{production} + 0.25 \cdot \text{diversity} + 0.20 \cdot \text{expansionAccess} + 0.15 \cdot \text{portSynergy}.$

This rubric uses the shared `production`, `diversity`, `expansionAccess`, and `portSynergy` definitions from Sections A.2.1–A.2.4. Production receives the highest weight because early income determines the opening trajectory, while the other terms preserve resource mix, expansion geometry, and port upside. The evaluator applies the rubric above to the shared settlement-quality quantities and clamps the final score to  $[0, 1]$ .

The threshold for `initial-settlement-location-selection` is 0.75 because opening settlement quality is highly consequential but multiple strong starts can be strategically defensible.

#### A.3.2 SETTLEMENT-LOCATION-SELECTION

This task uses the same scoring system as the initial-settlement task in Section A.3.1. The core difference between initial and subsequent location selection is that the initial settlement location selection task has a much higher weightage in the overall benchmark. This reflects the fact that the first settlement sets the foundation for the entire game and is a critical decision point, while subsequent settlement placements, although still important, are often more constrained by the existing board state and may have less impact on the overall game trajectory.

The threshold for `settlement-location-selection` is 0.90 because later settlement choices are evaluated against a smaller legal option set and should closely match the best available site.

#### A.3.3 ROAD-PLACEMENT-DIRECTION

$$\begin{aligned} \text{Score} = & 0.50 \cdot \text{futureReachability} + 0.35 \cdot \text{reachableValue} \\ & + 0.15 \cdot \text{blockingValue}. \end{aligned}$$

Where:

- `futureReachability` is the forward vertex’s onward legal-road count, clamped after division by 3.
- `reachableValue` is the forward vertex’s weighted settlement value:  
 $\text{reachableValue} = 0.5 \cdot \text{production} + 0.3 \cdot \text{diversity} + 0.2 \cdot \text{portSynergy}.$
- `blockingValue` is the maximum blocking score over the candidate road’s two endpoints and all opponents present at those endpoints.

For each endpoint-opponent pair, the evaluator ignores endpoints occupied by the acting player and endpoints occupied by a different opponent. It then computes:

$$O_{\text{branch}} = \text{open incident edges excluding the candidate road},$$

$$\text{branchScarcity} = \text{clamp} \left( \frac{2 - O_{\text{branch}}}{2} \right),$$

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$$\text{vertexOpportunity} = \begin{cases} 0.8, & \text{if the opponent already owns the endpoint,} \\ 1.0, & \text{if the endpoint is an open legal settlement site,} \\ 0.35, & \text{otherwise} \end{cases}$$

Let  $R_{\text{adj}}$  be the number of opponent roads adjacent to the endpoint,  $B_{\text{opp}}$  be 1 if the opponent owns the endpoint and 0 otherwise,  $L_{\text{opp}}$  be the opponent's road length, and  $H_{\text{opp}}$  be 1 if the opponent has Longest Road and 0 otherwise.

$$\begin{aligned} \text{presenceStrength} &= \text{clamp}\left(\frac{R_{\text{adj}} + B_{\text{opp}}}{2}\right), \\ \text{roadPressure} &= \text{clamp}\left(\frac{L_{\text{opp}} + 2 \cdot H_{\text{opp}}}{8}\right). \end{aligned}$$

The candidate blocking score is

$$\begin{aligned} \text{candidateBlocking} &= \text{clamp}(0.50 \cdot \text{branchScarcity} + 0.30 \cdot \text{vertexOpportunity} \\ &\quad + 0.10 \cdot \text{presenceStrength} + 0.10 \cdot \text{roadPressure}). \end{aligned}$$

and  $\text{blockingValue}$  is the largest  $\text{candidateBlocking}$ , or 0 if no opponent is present at either endpoint. The current implementation therefore recognizes common denial plays without running a full tactical search. Future reachability is weighted highest because roads are primarily investment moves. Reachable intersection value matters because roads derive value from the future settlement points they unlock. Blocking value is tactical and useful, but not every road is a denial move.

The threshold for `road-placement-direction` is 0.80 because roads are path-dependent investment moves and several near-best routes can preserve similar future options.

#### A.3.4 KNIGHT-CARD-PLAYING-DECISION

Define:

- `robberThreatLevel` measures how costly the current robber placement is for the player.
- `resourceNeed` is the largest entry in the goal-aware resource-need map from Section A.2.6.
- `armyProgressRatio` measures progress toward Largest Army after playing the Knight:

$$\begin{aligned} \text{robberThreatLevel} &= \begin{cases} \text{clamp}(\text{robberNumberWeight}/5), & \text{if the robber blocks the player} \\ 0.15, & \text{otherwise} \end{cases} \\ \text{resourceNeed} &= \max_r \text{need}[r]. \end{aligned}$$

This captures how valuable a Knight steal is when the player is missing an important resource for a near-term plan. Let  $K_{\text{self}}$  be the player's current Knight count and  $K_{\text{opp}}$  be the largest opponent Knight count. Then:

$$\text{armyProgressRatio} = \text{clamp}\left(\frac{K_{\text{self}} + 1}{\max(3, K_{\text{opp}} + 1)}\right).$$

Playing a Knight is most valuable when the robber is harming the player, when stealing a needed resource is likely to matter, and when Largest Army progress matters. The score does not include resource hand size because the Knight card has already been purchased and playing it does not reduce seven-risk for the player's resource hand.

The live scores are

$$\begin{aligned} \text{playNowScore} &= 0.45 \cdot \text{robberThreatLevel} + 0.35 \cdot \text{resourceNeed} \\ &\quad + 0.20 \cdot \text{armyProgressRatio} \\ \text{holdScore} &= \text{clamp}(0.25 + 0.75 \cdot (1 - \text{playNowScore})). \end{aligned}$$

The threshold for `knight-card-playing-decision` is 0.90 because playing a Knight should be reserved for clearly valuable robber relief, resource need, or Largest Army progress.



### A.3.5 BUILD-VS-SAVE-DECISION

This decision compares the expected value of spending now against preserving option value for later, incorporating immediate gains, future unlock potential, and seven-risk. The runtime constructs plan candidates for road, settlement, city, and devCard. Each plan uses the shared baseScore definitions from Section A.2.5, a cost vector, and a missingCount. missingCount is the total number of missing resource cards for the plan's cost vector. The discounted future value is

$$\text{effectiveScore} = \frac{\text{baseScore}}{1 + 0.75 \cdot \text{missingCount}}.$$

Each affordable build option gets  $\text{score} = \text{clamp}(\text{baseScore})$ . The save alternative is

$$B_{\text{eff}} = \text{best unaffordable effectiveScore}.$$

$$\text{saveScore} = \text{clamp}\left(\max(0.35, B_{\text{eff}} + S_7)\right),$$

where  $S_7$  is 0.1 when  $\text{resourceTotal} > 7$  and 0 otherwise.

The threshold for build-vs-save-decision is 0.90 because spend/save choices compare broad strategic options and should be close to the best current plan.

### A.3.6 YEAR-OF-PLENTY-CARD-PLAYING-DECISION

Year of Plenty is strongest when it immediately unlocks a valuable build, repairs bottlenecks, and improves future resource flexibility. The runtime waits until both chosen resources are known, then scores every unordered resource pair using the shared plan table from Section A.2.5. It computes beforeBest as the largest baseScore affordable before the card, or 0 if no plan is affordable. It then simulates each candidate resource pair and computes afterBest the same way from the simulated post-card hand. For each pair:

$$\text{buildUnlock} = \text{clamp}(\max(0, \text{afterBest} - \text{beforeBest}) + U_{\text{imm}}),$$

where  $U_{\text{imm}}$  is afterBest if afterBest > beforeBest and 0 otherwise.

$$\text{resourceScarcity} = \text{clamp}\left(\frac{\text{need}[r_1] + \text{need}[r_2]}{2}\right),$$

where  $r_1$  and  $r_2$  are the two resources in the candidate pair.

$$\text{longtermDiversityGain} = \text{clamp}\left(\frac{\text{distinctAfter} - \text{distinctBefore}}{2}\right).$$

The final score is

$$\text{Score} = \text{clamp}(0.50 \cdot \text{buildUnlock} + 0.35 \cdot \text{resourceScarcity} + 0.15 \cdot \text{longtermDiversityGain}).$$

The threshold for year-of-plenty-card-playing-decision is 0.90 because Year of Plenty should usually select resources that materially improve a near-term plan.

### A.3.7 MONOPOLY-CARD-PLAYING-DECISION

Monopoly is strongest when opponents likely hold a concentrated resource, the acting player currently needs that resource, and the resulting swing unlocks a build. For each candidate called resource, the server sums opponents' actual holdings of that resource, computes beforeBest, simulates adding all stolen copies, and recomputes afterBest. It then uses:

$$\text{estimatedOpponentHoldings} = \text{clamp}\left(\frac{C_r}{C_{\text{max}}}\right),$$

where  $C_r$  is the total opponent copies of the called resource and  $C_{\text{max}}$  is the largest opponent holding count across any resource.

$$\text{personalResourceNeed} = \text{clamp}(\text{need}[r]).$$

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486  $\text{buildUnlock} = \text{clamp}(\max(0, \text{afterBest} - \text{beforeBest}) + U_{\text{imm}}),$   
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 488 where  $U_{\text{imm}}$  is  $\text{afterBest}$  if  $\text{afterBest} > \text{beforeBest}$  and 0 otherwise. The resource score  
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 490  $\text{resourceScore} = \text{clamp}(0.55 \cdot \text{estimatedOpponentHoldings} + 0.30 \cdot \text{personalResourceNeed}$   
 491  $\quad + 0.15 \cdot \text{buildUnlock}).$   
 492  
 493 Monopoly plays emit a benchmark payload over the five possible resource calls. The implementa-  
 494 tion uses actual hidden-opponent holdings available on the server.  
 495 The threshold for `monopoly-card-playing-decision` is 0.90 because Monopoly calls  
 496 should closely match the strongest resource concentration and personal need signal.  
 497  
 498 **A.3.8 ROAD-BUILDING-CARD-PLAYING-DECISION**  
 499  
 500 The central question is whether playing Road Building now is more valuable than holding it for a  
 501 better timing window. The evaluator scores every currently legal road placement using the road-  
 502 placement heuristic from Section A.3.3, sorts scores descending, averages the best two legal road  
 503 scores (or one if only one exists), and computes  
 504  $\text{playNowScore} = \text{clamp}(0.70 \cdot \text{topTwoAverage} + 0.30 \cdot \text{longestRoadLeverage}),$   
 505  
 506 where  
 507 
$$\text{longestRoadLeverage} = \text{clamp}\left(\frac{(L_{\text{self}} + 2) - L_{\text{opp}} + 3}{6}\right).$$
  
 508  
 509 Here  $L_{\text{self}}$  is the player's current road length and  $L_{\text{opp}}$  is the best opponent road length. The hold  
 510 alternative is  
 511 
$$\text{holdScore} = \text{clamp}(0.30 + 0.70 \cdot (1 - \text{playNowScore})).$$
  
 512  
 513  
 514 The threshold for `road-building-card-playing-decision` is 0.85 because Road Build-  
 515 ing timing has tactical flexibility and a slightly wider range of good play-now choices.  
 516  
 517 **A.3.9 ROBBER-VICTIM-SELECTION**  
 518  
 519 This task is evaluated as an option ratio over legal victims. Victim selection is a discrete choice  
 520 whose value depends on leader pressure, theft value, and visible board strength. The visible leader  
 521 is the player with the highest public score:  
 522  
 523  $\text{visibleLeaderScore} = \text{visibleVP} + 2 \cdot \text{hasLongestRoad} + 2 \cdot \text{hasLargestArmy}.$   
 524  
 525 If multiple players tie, the implementation keeps the first tied player in turn-order/index order. For  
 526 each legal victim  $v$ , the score is  
 527  $\text{victimScore}(v) = \text{clamp}(\text{leaderBase}(v) + \text{handValue}(v) + \text{visibleStrength}(v)),$   
 528  
 529 where  
 530 
$$\text{leaderBase}(v) = \begin{cases} 0.55, & \text{if } v \text{ is the visible leader,} \\ 0.25, & \text{otherwise,} \end{cases}$$
  
 531  
 532 
$$\text{handValue}(v) = \min\left(0.35, \frac{\text{resourceTotal}(v)}{20}\right),$$
  
 533  
 534 
$$\text{visibleStrength}(v) = \min\left(0.10, \frac{\text{visibleScore}(v)}{20}\right).$$
  
 535  
 536  
 537  
 538 The threshold for `robber-victim-selection` is 0.90 because victim choice should usually  
 539 target the best blend of leader pressure and theft value.

### A.3.10 ROBBER-PLACEMENT

$$\text{Score} = 0.35 \cdot \text{hurtLeader} + 0.30 \cdot \text{productionDamage} + 0.20 \cdot \text{theftValue} + 0.15 \cdot \text{selfHarmAvoidance}.$$

For a candidate hex  $h$ , the evaluator uses the same visible leader definition from Section A.3.9. It identifies all opponents on that hex and computes each opponent's strength proxy:

$$\text{strength}(p) = \min \left( 1, \frac{\text{visibleVP}(p) + \text{resourceTotal}(p)/8}{5} \right).$$

Let  $w(h)$  be the dice-number probability weight of the hex. The damage terms are:

$$\text{hurtLeader} = \min \left( 1, \frac{w(h) \sum_{p \in h, p = \text{visible leader}} \text{strength}(p)}{3} \right),$$

$$\text{productionDamage} = \min \left( 1, \frac{w(h) \sum_{p \in h, p \neq \text{self}} \text{strength}(p)}{5} \right),$$

$$\text{theftValue} = \max_{p \in h, p \neq \text{self}} \min \left( 1, \frac{\text{resourceTotal}(p)}{7} \right),$$

$$\text{selfHarmAvoidance} = \begin{cases} 0, & \text{if the acting player has a building on } h, \\ 1, & \text{otherwise.} \end{cases}$$

Hurting the leader is the anti-runaway term, production damage captures future income suppression, theft value captures immediate swing, and self-harm avoidance prevents obvious placements on the acting player's own production.

The threshold for `robber-placement` is 0.90 because robber placement is a direct tactical action and should strongly align with leader damage, production denial, and theft value.

### A.3.11 ACCEPT-OR-REJECT-TRADE-OFFERS

$$\text{targetAcceptScore} = 0.50 \cdot \text{selfGain} + 0.25 \cdot \text{leaderPenalty} + 0.15 \cdot \text{resourceUrgency} + 0.10 \cdot \text{fairnessBalance}.$$

The current implementation approximates this as a two-option ratio over `accept` and `reject`. The accept score is the responder's simplified self-gain:

$$\text{acceptScore} = \text{selfGain} = \text{requestValue} - 0.5 \cdot \text{giveCost},$$

where `requestValue` uses the responder's need map and `giveCost` uses the responder's surplus map. The reject score is

$$\text{rejectScore} = \text{clamp}(1 - \text{acceptScore} + 0.1).$$

The threshold for `accept-or-reject-trade-offers` is 0.95 because accepting or rejecting a concrete offer should be nearly optimal to avoid rewarding harmful trades.

### A.3.12 GENERATE-TRADE-OFFERS

$$\text{tradeOfferScore} = 0.35 \cdot \text{legality} + 0.45 \cdot \text{selfGain} + 0.15 \cdot (1 - \text{leaderPenalty}) + 0.05 \cdot (1 - \text{fairnessPenalty}).$$

A trade must first be legal. Its main purpose is to improve the acting player's position. Helping the leader is strategically risky, and highly unfair offers are less plausible and less likely to be accepted. The components are:

$$\begin{aligned} \text{selfGain} &= \text{clamp}(\text{requestValue} - 0.5 \cdot \text{giveCost}), \\ \text{fairnessPenalty} &= \min(1, \max(0, \text{giveCost} - \text{requestValue})), \\ \text{legality} &= \begin{cases} 0, & \text{if the offer gives resources the player does not own,} \\ 1, & \text{otherwise,} \end{cases} \\ \text{leaderPenalty} &= \begin{cases} 1, & \text{if the target is the visible leader,} \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

The generated offer is compared against the benchmark-best score via a ratio.

The threshold for `generate-trade-offers` is 0.70 because generated offers are exploratory and many legal, plausible offers may be useful even if not benchmark-best.

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### A.3.13 SELECT-TARGETED-TRADE-PARTNER

This task is evaluated as an option ratio. Each candidate partner is scored as

$$\text{partnerScore} = \text{selfGain} + \text{availabilityBonus} - 0.4 \cdot \text{leaderHelpPenalty},$$

where

$$\text{availabilityBonus} = \begin{cases} 0.25, & \text{if the partner can provide the requested resources,} \\ 0, & \text{otherwise,} \end{cases}$$

$$\text{leaderHelpPenalty} = \begin{cases} 1, & \text{if the partner is the visible leader,} \\ 0, & \text{otherwise.} \end{cases}$$

This favors partners that produce good value and can satisfy the request, while discounting trades with the leader. Targeted trade proposals emit this task at runtime, and the selected target is compared against all opponent candidates rather than only against itself.

The threshold for `select-targeted-trade-partner` is 0.90 because targeted trade partner choice should closely match availability, self-gain, and leader-avoidance signals.

### A.3.14 GENERATE-COUNTER-TRADE-OFFER

$$\begin{aligned} \text{counterTradeScore} = & 0.35 \cdot \text{legality} + 0.45 \cdot \text{selfGain} \\ & + 0.15 \cdot (1 - \text{leaderPenalty}) + 0.05 \cdot (1 - \text{fairnessPenalty}). \end{aligned}$$

The structural reasoning follows the generated-trade-offer task from Section A.3.12, but now from the responder's perspective relative to an incoming offer. The payload reuses the trade-quality sub-scores from Section A.3.12:

$$\begin{aligned} \text{selfGain} &= \text{clamp}(\text{requestValue} - 0.5 \cdot \text{giveCost}), \\ \text{fairnessPenalty} &= \min(1, \max(0, \text{giveCost} - \text{requestValue})), \\ \text{legality} &= \begin{cases} 0, & \text{if the responder gives resources they do not own,} \\ 1, & \text{otherwise,} \end{cases} \\ \text{leaderPenalty} &= \begin{cases} 1, & \text{if the original proposer is the visible leader,} \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

The benchmark payload stores this through `offerContext` with `bestOfferScore = 1` rather than logging each candidate counteroffer separately.

The threshold for `generate-counter-trade-offer` is 0.70 because counteroffers are exploratory like generated offers and should allow a broader set of reasonable proposals.

### A.3.15 CITY-VS-SETTLEMENT-VS-ROAD-PRIORITIZATION

This task is evaluated as an option ratio. It compares build classes rather than board coordinates and asks which legal build family has the highest strategic value at the current state. Let the maxima below range over the currently legal options:

$$\text{settlementScore} = \max \text{settlementScore}(v),$$

$$\text{roadScore} = \max \text{roadScore}(e),$$

where settlement scores use Section A.3.2 and road scores use Section A.3.3.

$$\text{cityScore} = 0.65 \cdot \text{production} + 0.10 \cdot \text{diversity} + 0.05 \cdot \text{portSynergy} + 0.20.$$

Production dominates because a city doubles an existing production site; diversity and port access still matter because they determine how valuable that amplification is, and the constant 0.20 reflects the guaranteed VP and output increase. The task is emitted for settlement, city, and normal road-building choices. Free roads from Road Building are intentionally excluded.

The threshold for `city-vs-settlement-vs-road-prioritization` is 0.90 because build-family prioritization should usually select the strongest currently legal development path.

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### A.3.16 DEVELOPMENT-CARD-PURCHASE-DECISION

This task is evaluated as an option ratio. Buying a development card is an investment choice against competing uses of resources and should be favored when hidden VP, Knight access, or tactical flexibility outweigh immediate board development. This decision is derived from the shared `devCard` plan `baseScore` from Section A.2.5:

$$\text{devCardBaseScore} = \max(0.2, 0.55 \cdot (0.5 + 0.5 \cdot \text{cityPlanScore}))$$

if the development deck is available, otherwise 0. The two options are

$$\text{buy-dev-card} = \text{clamp}(\text{devCardBaseScore},$$

$$\text{skip-dev-card} = \text{clamp}(\max(\text{saveScore}, B_{\text{nondev}})),$$

where  $B_{\text{nondev}}$  is the best affordable non-development-card plan `baseScore`. `buyDevCard` emits this task directly. If development-card purchase was affordable but skipped for the full turn, the negative case is recorded at turn end.

The threshold for `development-card-purchase-decision` is 0.95 because buying a development card is a precise resource-spending decision with clear competing alternatives.

### A.3.17 DISCARD-STRATEGY-AFTER-SEVEN

This task is scored as

$$\frac{\text{RetainedHandValue}}{\text{BestRetainedValue}}.$$

Discarding is constrained by a fixed number of cards to lose, and the actual strategic objective is to maximize the value of the retained hand rather than the discarded hand. The evaluator enumerates all legal discard bundles summing to  $\lfloor \text{total\_hand\_size}/2 \rfloor$ , subtracts each from the current hand, and scores the retained hand with:

$$\text{bestAffordable} = \max \text{ affordable plan baseScore},$$

$$\text{bestFuture} = \max \text{ plan effectiveScore},$$

$$\text{diversity} = \text{clamp}\left(\frac{\text{distinctPositiveResources}}{5}\right).$$

$$\text{retainedHandValue} = \text{clamp}(0.55 \cdot \text{bestAffordable} + 0.30 \cdot \text{bestFuture} + 0.15 \cdot \text{diversity}).$$

The final task score is the selected retained-hand value divided by the best legal retained-hand value. Candidate discard bundles are generated directly from the player hand and required discard count. The threshold for `discard-strategy-after-seven` is 0.95 because discard choices are fully enumerated and should preserve nearly the best retained-hand value.

### A.3.18 BANK-TRADE-DECISION

$$0.50 \cdot \text{resourceValueDelta} + 0.30 \cdot \text{buildUnlock} + 0.20 \cdot \text{opportunityCost}.$$

This task is currently approximated through a choice ratio over candidate bank/port trades plus a `no-trade` option. For each candidate trade, the evaluator computes the trade ratio using `GameLogic.getTradeRatio` and scores the candidate as

$$\text{tradeScore} = \text{clamp}(\text{need}[\text{get}] - 0.5 \cdot \text{surplus}[\text{give}] + 0.5),$$

and includes a fixed baseline

$$\text{no-trade} = 0.45.$$

Acquiring a needed resource is valuable, paying with a surplus resource is less painful than paying with a scarce one, and the `no-trade` baseline prevents weak legal trades from looking attractive by default. The named rubric components are not yet separately logged; they are compressed into a single trade desirability heuristic.

The threshold for `bank-trade-decision` is 0.85 because bank trades are deterministic but may have several near-equivalent resource-conversion options.

### A.3.19 DECIDE-PURSUE-LONGEST-ROAD

$$\text{legacyScore} = 0.40 \cdot \text{roadProgressRatio} + 0.30 \cdot \text{opponentRoadGap} \\ + 0.20 \cdot \text{resourceAffordability} + 0.10 \cdot \text{VPImpact}.$$

The stronger formulation is to record a pending task episode, wait 4–6 turns for the evaluated player, and score using snapshot-plus-outcome evidence:

$$\text{pursueScore} = 0.45 \cdot \text{blockingEffectiveness} + 0.30 \cdot \text{opponentProgressThreat} \\ + 0.15 \cdot \text{selfRoadSynergy} + 0.10 \cdot \text{resourceCost}.$$

This redesign is preferable because Longest Road decisions are not fully observable from a single action. The live system already uses the delayed v2-style episode. At decision time it snapshots player road length, strongest opponent road rival and their road length, whether either player currently holds Longest Road, player visible score, and roads built so far in benchmark long-term stats.

After a four-turn horizon, decisive resolution, or game end, it computes:

$$G_{\text{self}} = \max(0, L_{\text{self},\text{end}} - L_{\text{self},\text{start}}), \quad G_{\text{target}} = \max(0, L_{\text{target},\text{end}} - L_{\text{target},\text{start}}), \\ R_{\Delta} = \max(0, \text{roadsBuiltSinceSnapshot}), \\ \text{opponentProgressThreat} = \text{clamp} \left( \frac{(L_{\text{target},\text{start}} - L_{\text{self},\text{start}}) + 2H_{\text{target},\text{start}} + 2}{8} \right), \\ \text{blockingEffectiveness} = \text{clamp}(0.45 + 0.18G_{\text{self}} - 0.12G_{\text{target}} + 0.25H_{\text{self},\text{end}}), \\ \text{selfRoadSynergy} = \text{clamp} \left( \frac{G_{\text{self}}}{4} + 0.35H_{\text{self},\text{end}} \right).$$

Here each  $H$  indicator is 1 when that player has Longest Road at the named time and 0 otherwise. The resource-cost term is

$$\text{resourceCost} = \begin{cases} 1, & \text{successful defer and } R_{\Delta} = 0, \\ 0.5, & \text{pursue and } R_{\Delta} = 0, \\ \text{clamp} \left( \frac{G_{\text{self}} + \text{longestRoadBonus}}{R_{\Delta}} \right), & \text{otherwise.} \end{cases}$$

The final alternatives are

$$\text{pursueScore} = \text{clamp}(0.45 \cdot \text{blockingEffectiveness} + 0.30 \cdot \text{opponentProgressThreat} \\ + 0.15 \cdot \text{selfRoadSynergy} + 0.10 \cdot \text{resourceCost}), \\ \text{deferScore} = \text{clamp}(0.45 \cdot (1 - \text{opponentProgressThreat}) \\ + 0.25 \cdot \left( 1 - \min \left( 1, \frac{G_{\text{target}}}{3} \right) \right) \\ + 0.15 \cdot \left( 1 - \min \left( 1, \frac{G_{\text{self}}}{3} \right) \right) + 0.15D_0).$$

Here  $D_0$  is 1 if  $R_{\Delta} = 0$  and 0.2 otherwise. This task now runs as a pending long-horizon episode, and finalization writes one ordinary benchmark task result after the horizon, decisive resolution, or game end.

The threshold for decide-pursue-longest-road is 0.95 because long-horizon race commitments are scored after an outcome window and should reflect strong evidence.

### A.3.20 DECIDE-PURSUE-LARGEST-ARMY

$$\text{pursueScore} = 0.40 \cdot \text{knightCardRatio} + 0.30 \cdot \text{opponentArmyGap} \\ + 0.20 \cdot \text{devCardAffordability} + 0.10 \cdot \text{VPImpact}.$$

The rubric stated at the start of Section A.3.20 remains appropriate, but it should be evaluated over a 4–6 turn future window rather than immediately. Largest Army is not about one development-card purchase or one Knight; it is about whether a multi-turn commitment improves the player's race standing without unacceptable economic damage. A hybrid intent-plus-outcome model is appropriate because pure outcome scoring would overreact to deck order and dice luck, while pure immediate

scoring would ignore whether the strategic plan materialized. Like Longest Road, this task is implemented as a pending long-horizon episode. At decision time it snapshots player knights played, strongest opponent army rival and their knights played, whether either player currently has Largest Army, player visible score, player hand size, and development cards bought so far in benchmark long-term stats.

At finalization it computes:

$$C_{\Delta} = \max(0, \text{devCardsBoughtSinceSnapshot}), \quad K_{\Delta} = \max(0, K_{\text{self},\text{end}} - K_{\text{self},\text{start}}),$$

$$G_{\text{start}} = K_{\text{self},\text{start}} - K_{\text{target},\text{start}}, \quad G_{\text{end}} = K_{\text{self},\text{end}} - K_{\text{target},\text{end}},$$

$$\text{knightCardRatio} = \begin{cases} 1, & K_{\Delta} > 0 \text{ and } C_{\Delta} = 0, \\ 0, & K_{\Delta} = 0, \\ \text{clamp}\left(\frac{K_{\Delta}}{C_{\Delta}}\right), & \text{otherwise,} \end{cases}$$

$$\text{opponentArmyGap} = \text{clamp}\left(\frac{(G_{\text{end}} - G_{\text{start}}) + 3}{6}\right),$$

$$\Delta R = \text{currentHandSize} - \text{snapshotHandSize},$$

$$\text{devCardAffordability} = \text{clamp}\left(0.6 + \frac{\Delta R}{10} - 0.08C_{\Delta}\right),$$

$$\text{vpImpact} = \text{clamp}\left(\frac{\Delta \text{visibleVP} + H_{\text{army},\text{end}}}{3}\right).$$

Here  $H_{\text{army},\text{end}}$  is 1 if the player has Largest Army at finalization and 0 otherwise. The final alternatives are

$$\begin{aligned} \text{pursueScore} = & \text{clamp}(0.40 \cdot \text{knightCardRatio} + 0.30 \cdot \text{opponentArmyGap} \\ & + 0.20 \cdot \text{devCardAffordability} + 0.10 \cdot \text{vpImpact}), \end{aligned}$$

$$\begin{aligned} \text{deferScore} = & \text{clamp}(0.35 \cdot (1 - \text{opponentArmyGap}) + 0.25 \cdot (1 - \text{knightCardRatio}) \\ & + 0.20 \cdot R_{\text{safe}} + 0.20 \cdot (1 - H_{\text{army},\text{end}})). \end{aligned}$$

Here  $R_{\text{safe}}$  is 1 if  $\Delta R \geq 0$  and 0.4 otherwise.

The threshold for `decide-pursue-largest-army` is 0.95 because Largest Army pursuit is a long-horizon commitment and should be supported by strong snapshot-plus-outcome evidence.